

TIODIN

CONTENT

Each **Tiodin** tablet contains ticlopidine hydrochloride 250 mg

DESCRIPTION

Yellowish – white round biconvex film coated tablet.

ACTIONS

Ticlopidine hydrochloride is an inhibitor of platelet aggregation. Ticlopidine hydrochloride interferes with platelet membrane function by inhibiting adenosine diphosphate (ADP)- induced binding of fibrinogen to the platelet membrane at a specific receptor site (the glycoprotein lib-IIIa complex). Subsequently, release of platelet granule constituents, platelet-platelet interactions, and platelet adhesion to the endothelium and to atheromatous plaque are inhibited.

Ticlopidine hydrochloride causes an irreversible effect on platelet function for the life of the platelet, and also prolongation of bleeding time.

PHARMACOLOGY

Pharmacodynamics:-

Clinically significant inhibition (more than 50%) of platelet aggregation occurs within 4 days after repeated dosing of 250mg twice a day. Maximum inhibition of platelet aggregation (60 to 70%) is achieved in 8 to 11 days. Doses higher than 500mg per day do not produce a significant additional increase in the extent of inhibition.

In the majority of patients, bleeding time and other platelet function tests return to normal levels within 1 to 2 weeks after discontinuation of ticlopidine hydrochloride.

At the recommended dose of 250mg twice daily, ticlopidine hydrochloride has no known significant pharmacological actions other than inhibition of platelet function and prolongation of bleeding time.

Pharmacokinetics

Ticlopidine hydrochloride is rapidly (80% or more of a dose) absorbed from the gastro-intestinal tract. Absorption is increased when medication is taken after a meal. Time to reach peak concentration after a single 250mg dose is about 2 hours. The elimination half-life of ticlopidine hydrochloride appears to be longer in older subjects than in younger subjects.

Ticlopidine hydrochloride is extensively bound to plasma protein (90%), primarily to serum albumin and lipoprotein and to a lesser extent (15% or less) to alpha-1-acid glycoprotein.

Ticlopidine hydrochloride undergoes extensive hepatic metabolism. Unchanged ticlopidine hydrochloride accounts for only trace amounts of the quantity eliminated in the urine.

Plasma clearance rate decreases in elderly patients (average age 70 years) as well as in renally impaired patients.

TOXICOLOGY

No evidence of carcinogenicity or tumorigenicity and mutagenicity has been found in animal studies.

INDICATIONS

Reducing risk of thrombotic stroke (fatal or non fatal) in patients who have experienced stroke precursors and in patients who have had a complete stroke.

CONTRAINDICATIONS

In blood diseases or haemorrhagic diathesis where bleeding time is prolonged. It is similarly contraindicated in case of

organic lesions with a tendency to bleed: Gastro duodenal ulcer in periods of activity during the acute phase of a haemorrhagic cerebrovascular accident or severe liver dysfunction. Ticlopidine is also contraindicated in patients who have history of allergy to ticlopidine, history of leukopenia, thrombocytopenia or agranulocytosis, and in patients with severe hepatic impairment, since they may be at risk for bleeding diathesis.

SIDE EFFECTS / ADVERSE REACTIONS

Most of the side effects reported with ticlopidine include neutropenia or agranulocytosis, thrombocytopenia, gastro-intestinal disturbances and skin rash, appear within the first 3 months of treatment, although some may occur or recur several months later. Rarely, neutropenia, thrombocytopenia or thrombotic thrombocytopenia purpura has occurred after years of treatment. Fatalities associated with severe neutropenia or thrombotic thrombocytopenia purpura have been reported.

Ticlopidine-induced GI disturbances may occur in up to 40% of the patients receiving the medication. They are generally mild and usually disappear within 1 or 2 weeks without discontinuation of treatment.

Ticlopidine has been reported to cause increased bleeding, spontaneous posttraumatic bleeding and preoperative bleeding including, but not limited to gastrointestinal bleeding. It has also been associated with bleeding complications such as conjunctival haemorrhage, epistaxis, haematuria and ecchymosis.

Ticlopidine may cause altered laboratory findings as follows:

a) Hematological

Neutropenia and rarely, pancytopenia as well as isolated thrombocytopenia or exceptionally accompanied by haemolytic anemia, have been associated with ticlopidine treatment.

b) Hepatic

Ticlopidine treatment has been accompanied by an increase in hepatic enzymes. Common increase (either isolated or not) of alkaline phosphatases and transaminases (incidence greater than twice the upper limit of normal) was observed in both groups (ticlopidine and placebo). Ticlopidine treatment has also been accompanied by a minor elevation of bilirubin.

c) Cholesterol

Chronic ticlopidine therapy has been associated with increased serum cholesterol and triglyceride levels. Serum HDL-C, LDL-C, VLDL-C and triglyceride levels may increase 8 to 10% after 1 to 4 months of treatment. No further elevations are seen with continued therapy. The ratios of lipoprotein subfractions (especially the ratio of HDL to LDL) remain unchanged. The data in the clinical studies have shown that the effect does not depend on age, sex, alcohol consumption or diabetes, and has no influence on the cardiovascular risk.

Other potential side effects, although rare, may include:

- hepatitis or cholestatic jaundice
- common skin rashes, particularly maculopapular or urticarial, often accompanied by pruritus. In general, skin rashes occur within the first 3 months of treatment, with a mean time to onset of 11 days. If treatment is discontinued, the symptoms disappear within a few days. These skin rashes may be generalized.
- Immunological reactions with different manifestations such as Quincke edema, vasculitis, lupus syndrome, hypersensitivity and nephropathy.
- pancytopenia, haemolytic anaemia with reticulocytosis, immune thrombocytopenia
- arthropathy and myositis
- hyponatremia

PRECAUTIONS / WARNINGS

Because of the risk of ticlopidine-induced neutropenia and/or thrombocytopenia, which may be life threatening, ticlopidine should be reserved for patients who are unable to tolerate aspirin therapy indicated in stroke prevention.

As there is a risk of neutropenia and thrombocytopenia, it is important to monitor complete blood counts with differential and platelet counts to detect potential adverse effects at the beginning of therapy and every 2 weeks for the first 3 months of treatment.

As ticlopidine has long plasma half-life, patients who discontinue ticlopidine during this 3 months period should continue to be monitored for 2 weeks after discontinuation.

Signs and symptoms which may be suggestive of neutropenia include fever, pharyngitis, ulcers of buccal mucosa, and that of thrombocytopenia include prolonged bleeding, bruising, purpura, dark stool and dark urine. If any of these symptoms occur, patients should be advised to stop medication and consult the physician immediately. It is recommended to discontinue ticlopidine therapy temporarily if an injury will result in substantial risk of bleeding. Caution should be taken in patients who may be at risk of bleeding from trauma or pathological condition.

It is recommended to discontinue ticlopidine 10 to 14 days prior to elective surgery, including dental extraction because of the risk of increase blood loss. For dental extraction, ticlopidine may cause neutropenia, which may result in an increased incidence of microbial infection, delayed healing and gingival bleeding. If severe neutropenia occurs, dental work should be deferred until blood counts have returned to normal. In emergency elective surgical situation, transfusion of fresh platelets may improve haemostasis. Although intravenous administration of 20mg of methylprednisolone to ticlopidine-treated patients has been shown to return the bleeding time to normal within 2 hours, the effect of such treatment on perisurgical haemostasis has not been established.

Since ticlopidine is metabolised by the liver, it should be used with caution in patients with impaired liver function. Treatment should be discontinued if hepatitis or jaundice develops.

Decreased plasma clearance, increased AUC values and prolong bleeding times can occur in renally impaired patients. Therefore, it may be necessary to reduce the dosage of ticlopidine or discontinue it altogether.

Thrombotic Thrombocytopenia Purpura (TTP)

TTP is characterised by thrombocytopenia, microangiopathic haemolytic anaemia (schistocytes seen on peripheral smear), neurological findings, renal dysfunction and fever. TTP might also be suggested by symptoms such as weakness, pallor, fever, jaundice or purpura.

Use in Pregnancy and Lactation

Adequate and well-controlled studies have not been performed in pregnant women. It is not known whether ticlopidine is distributed into human breast milk. Therefore, safe use has not been established.

Geriatrics

Appropriate studies performed have not demonstrated geriatrics-specific problems that would limit the usefulness of ticlopidine in the elderly. No overall differences in efficacy or safety were observed between elderly and younger patients. However, it should be noted that clearance of ticlopidine is somewhat lower in elderly patients and that the trough levels are increased.

Paediatrics

No information is available on the relationship of age, to the effects of ticlopidine in paediatric patients. Safety and efficacy have not been established in this group of patients.

Drug Interaction

Concurrent administration of NSAIDs, salicylic derivatives (aspirin), antiplatelets drugs, oral anticoagulants, anticoagulants, coumarin or indandione-derivative or heparin or thrombolytic agents such as alteplase, anistreplase, streptokinase, urokinase and ticlopidine may result in increased risk of bleeding. Plasma concentration of ticlopidine is decreased when administered after aluminium and magnesium containing antacid. Combinations requiring special precautions: theophylline (risk of overdose due to a decrease of plasma theophylline clearance), digoxin (15% decrease of plasma digoxin level not affecting its therapeutic efficacy). Concurrent use of aspirin and ticlopidine is not recommended. When xanthine is administered to a patient receiving ticlopidine, the half-life of xanthine may be increased.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION

Tiodin should be taken with meals to achieve maximum absorption and tolerance. Administration of **Tiodin** with food also reduces the risk of gastrointestinal side effects.

Adults above 18 years of age: orally one tablet (250mg), twice daily with food.

Geriatric patients: same as dosage in adult above 18 years old.

Children below 18 years old: safety and efficacy have not been established.

SYMPTOMS AND TREATMENT OF ADVERSE EFFECTS (OVERDOSAGE)

Symptoms of acute toxicity include gastrointestinal haemorrhage, convulsions, hypothermia, dyspnea, loss of equilibrium and abnormal gait.

In case of overdosage, administration of ticlopidine should be withdrawn immediately, followed by induced vomiting, gastric lavage and other general supportive measures.

For bleeding complications: Although administration of methylprednisolone (20 mg, intravenously) returns the bleeding time to normal in ticlopidine-treated patients, clinical experience indicating that such treatment improves haemostasis is lacking. Platelet transfusion may be helpful but since it may accelerate thrombosis in patients with TTP on ticlopidine, it should, if possible, be avoided. In addition, other measures to control bleeding in specific areas must be employed as needed.

PACKING / PACK SIZE

Aluminium foil strip of 10 tablets per strip, box of 10 X 10 tablets.

STORAGE CONDITIONS, USER INFORMATIONS AND PHARMACEUTICAL PRECAUTIONS

Store in a cool dry place at below 30 °C.

Protect from moisture, freezing, excessive heat & light.

Keep out of reach of children.

(Jauhi daripada kanak-kanak)

Shelf life: as indicated on the outer box.

MANUFACTURED BY

T.O. PHARMA CO.,LTD.

101 Ladprao 124 (Sawatdikan),
Ladprao Road, Phlapphla, Wangthonglang,
Bangkok 10310, Thailand.

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